

Anatomy and Physiology C - EXAM PACKET

UT Austin Invitational Tournament 2025

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_____ / 192 points



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Event Parameters:

- Each team may bring **one** 8.5" x 11" sheet of paper that may contain information on both sides with **two** stand-alone non-programmable, non-graphic calculators.
- There are **170 questions** on this exam with a total of **192 points**. Point values are denoted throughout the exam.
- There are **10** tiebreakers on this exam. They will be denoted as **TB#**.
- The exam authors do **not** take credit for any images used for this exam. Credits are attributed to the proper owners.
- Write your team number and answers in the "Answer Sheet" provided. Please do **not** write on the exam packets.
- Good luck and have fun! :)

I. Nervous System

[68 points]

T/F: Determine if the provided statement is **true (T)** or **false (F)**. Each is worth **one** point. (3)

1. The corpus callosum is primarily composed of white matter.
2. Most strokes are ischemic in nature- meaning their root cause is the blockage of an artery.
3. **TB#1** The Nernst and Goldman-Hodgkin-Katz equations both are rearrangements of the same fundamental relation between charge and permeability of multiple ionic species.

Multiple Choice: Choose the most appropriate answer for the following questions. Each question is worth **one** point. (15)

4. These are the most abundant glial cells found in the CNS.
 - a. Oligodendrocytes
 - b. Schwann cells
 - c. Satellite cells
 - d. Astrocytes
 - e. Ependymal cells
5. Node of Ranvier:
 - a. Are unmyelinated regions between two adjacent myelinated axon segments
 - b. Contains a small amount of Na⁺ channels
 - c. Decreases action potential velocity
 - d. A and B
 - e. All of the above
6. Which supporting glial cell transports substances between the blood and the ventricles and is mainly found in the floor of the third ventricle?
 - a. Astrocytes
 - b. Microglia
 - c. Ependymal cells
 - d. Tanycytes
 - e. Oligodendrocytes
7. In response to CNS tissue damage, which supporting glial cell is activated in response and proliferates, migrates to the damaged tissue, and releases inflammatory mediators?
 - a. Astrocytes
 - b. Microglia
 - c. Ependymal cells
 - d. Tanycytes
 - e. Oligodendrocytes
8. Myelin wraps around nerve axons in a spiral structure and acts as an insulator. Myelin also:
 - a. Increases space constant
 - b. Increases membrane resistance
 - c. Increases membrane capacitance
 - d. A and B
 - e. All of the above
9. Which of the following is a risk factor for cerebral palsy?
 - a. Preterm birth
 - b. Low birth weight
 - c. Perinatal asphyxia
 - d. A and B
 - e. All of the above

10. What peripheral nerve layer is the layer around nerve fascicles and contains the blood-nerve barrier?
- Endoneurium
 - Perineurium
 - Epineurium
 - None of the above
11. What is the conduction velocity of A-alpha fibers?
- 60-120 m/s
 - 2-30 m/s
 - 3-15 m/s
 - 0.25-1.5 m/s
12. What nerve fiber is not myelinated and responsible for the detection and transmission of pain?
- A-alpha fibers
 - A-beta fibers
 - A-delta fibers
 - B fibers
 - C fibers
13. Which of the following is an example of an unconventional neurotransmitter?
- GABA
 - Adenosine
 - Substance P
 - Neuropeptide Y
 - Nitric oxide
14. Which of the following neurotransmitters is mainly synthesized for glutamate via glutamate decarboxylase and is inhibitory?
- Aspartate
 - Dopamine
 - Endorphins
 - Enkephalins
 - GABA
15. What is the site of action of norepinephrine?
- Basal nucleus of Meynert
 - Substantia nigra
 - Locus coeruleus
 - Raphe nucleus
 - Nucleus accumbens
16. Increased levels of what neurotransmitter is associated with Parkinson disease?
- Acetylcholine
 - Dopamine
 - Norepinephrine
 - Serotonin
 - GABA
17. Decreased levels of what neurotransmitter is associated with Alzheimer disease?
- Acetylcholine
 - Dopamine
 - Norepinephrine
 - Serotonin
 - GABA

18. Which of the following statements is true regarding the etiology of multiple sclerosis?
- a. High vitamin D levels is an environmental risk factor of MS. HLA-DRB1*15 allele presence is associated with an increased risk of MS.
 - b. HLA-A*02 allele presence is associated with an increased risk of MS.
 - c. A and B
 - d. All of the above
19. **TB#2** What medication that is used for multiple sclerosis suppresses T cell activity and leads to decreased proinflammatory cytokines and decreased lymphocyte invasion of the CNS?
- a. Dimethyl fumarate
 - b. Teriflunomide
 - c. Glatiramer acetate
 - d. Interferon beta
 - e. Cladribine

Matching: Please refer to **image packet** (pg. 1). Select the label below which corresponds best to the description provided. Some labels may be used more than once, while some may not be used at all. Each match is worth **one** point. **(8)**

Resource: University of Michigan - BlueLink Photographic Atlas

20. Parietal lobe.
21. Frontal lobe.
22. Cerebellum.
23. Generally regarded as the main integration center of sensory information.
24. **TB#3** One's working memory and executive control are generally attributed to this lobe.
25. Degeneration of this structure is associated with memory-related changes in those with Alzheimer's.
26. Which label aside from **Label E** does NOT directly contain components of the limbic system?
27. The type of neurons affected by Amyotrophic Lateral Sclerosis (ALS) are largely concentrated within this region.

Fill-in: Complete the sentences with the most appropriate answer. Be as clear and concise as possible. Each is worth **one** point. **(6)**

28. _____ is the main glial cells in cerebral white matter and can myelinate 30 axons on average.
29. _____ promotes repair after nerve injury and myelinates axons of the PNS.
30. Acetylcholine is synthesized from _____ and acetyl coenzyme A.
31. Dopamine is both excitatory and inhibitory and involved in initiating _____.
32. G-protein-gated neurotransmitter receptors is an example of a _____-gated ion channel.
33. _____ synapses signal between axons.

Histology: Use the provided slides in the **image packet** (pg. 1) to answer the following questions. Point values will be denoted. **(6)**

Resource: University of Michigan - Histology

34. What name is used for the synapse in **Slide 1**?
35. What is the neurotransmitter used at this synapse?
36. What is the structure encircled by the red dashed line in **Slide 2**?
37. This structure in **Slide 2** is lined by what type of glial cell ?
38. What is the main product secreted by these glial cells into the ventricular spaces?
39. What are the cells that are stained black in **Slide 3**?

Classification: Determine if the following statements apply to **ionotropic** receptors or **metabotropic** receptors. If ionotropic, write **I** in your answer sheet. If metabotropic, write **M**. If both, write **B**. Each is worth **one** point. **(10)**

40. Fast receptor opening time
41. Signal transduction mechanisms such as G-proteins
42. Receptor and ion channel are separate molecules
43. Short-lasting postsynaptic potential
44. Nicotinic acetylcholine receptor

- 45. Muscarinic acetylcholine
- 46. Glutamate receptor
- 47. GABA_A receptor
- 48. Histamine receptor
- 49. Effects are more widespread and persistent throughout cell

Toxins: Answer the following questions regarding toxins and their effects on neuron function and signaling. Each is worth **one** point. **(5)**

- 50. Bacteria inhabiting the gut of puffer fish can produce a neurotoxin called?
 - a. Tetrodotoxin
 - b. Curare
 - c. Botulinum toxin
 - d. Anatoxin-a
 - e. Tetanus toxin
- 51. Which neurotoxin is also known as Very Fast Death Factor?
 - a. Tetrodotoxin
 - b. Curare
 - c. Botulinum toxin
 - d. Anatoxin-a
 - e. Tetanus toxin
- 52. Curare is a neurotoxin that has been used for centuries by American Indigenes and was known as “arrow poison.” It is a long-acting neuromuscular blocking agent that primarily competes with?
 - a. Dopamine
 - b. Serotonin
 - c. Acetylcholine
 - d. Nitric Oxide
 - e. Glycine
- 53. **T/F:** The Tetanus toxin acts as a protease and prevents the release of excitatory neurotransmitters from Renshaw cells in the spinal cord. **(1)**
- 54. **T/F:** The Botulinum neurotoxin is produced by *Clostridium botulinum* and blocks the release of GABA from the neuromuscular junction. **(1)**

Labeling: Label the diagram in the **image packet** (pg. 1). Each is worth **one** point. **(15)** *Reference: Konan et al. (2023).*

55. A

56. B

57. C

58. D

59. E

60. F

61. G

62. H

63. I

64. J

65. K

66. L

67. M

68. N

69. O

II. Sense Organs

[42 points]

T/F: Determine if the provided statement is **true (T)** or **false (F)**. Each is worth **one** point. (5)

- 70. The amino acid tryptophan is primarily responsible for the umami flavor of savory food such as meats or cheeses.
- 71. The facial (VII) nerve is responsible for collecting sensations of taste from the largest area of the tongue.
- 72. While most odors act via receptors and second messenger systems, some odorants such as ammonia, chlorine, and capsaicin act via the vagus (X) nerve instead.
- 73. **TB#4** Structures at the base of the cochlea are responsible for sensing low-pitched sounds, while structures at the apex of the cochlea are responsible for sensing high-pitched sounds.
- 74. Sound can be transmitted through a vacuum.

Multiple Choice: Choose the most appropriate answer for the following questions. Each question is worth **one** point. (14)

- | | |
|--|---|
| <p>75. Which of the following is not considered a special sense?</p> <ul style="list-style-type: none">a. Visionb. Hearingc. Equilibriumd. Stretche. Smell | <p>77. The bumps on one's tongue are best described as?</p> <ul style="list-style-type: none">a. Taste budsb. Basal cellsc. Supporting cellsd. Lingual papillaee. Taste hairs |
| <p>76. Taste buds can be found in all of the following locations with the exception of the:</p> <ul style="list-style-type: none">a. Tongueb. Pharynxc. Epiglottisd. Soft palatee. Hard palate | <p>78. The TRPV1 receptor, responsible for spicy sensations from molecules such as capsaicin, is best described as a:</p> <ul style="list-style-type: none">a. Thermoreceptorb. Photoreceptorc. Nociceptord. Mechanoreceptore. More than one of the above |

79. Most of the tongue's taste buds can be found on:
- Filiform papillae
 - Foliate papillae
 - Fungiform papillae
 - Vallate papillae
 - Taste buds are equally distributed across all these structures
80. The threshold for tasting _____ flavors is the lowest of all the tastes.
- Salty
 - Sour
 - Sweet
 - Umami
 - Bitter
81. Which of these structures is not found in the region considered the outer ear?
- Auricle
 - Cerumen
 - Pinna
 - Eustachian tube
 - Lobule
82. Hearing loss caused by severe otitis externa is generally classified as:
- Sensorimotor
 - Sensorineural
 - Conductive
 - A mix of the above
 - None of the above; otitis externa does not affect hearing
83. One may experience discomfort as a plane lifts off and the external air pressure drops. The subsequent pressure increase in the ear is relieved by opening of the:
- Auditory canal
 - Eustachian tube
 - Oval window
 - Round window
 - Nasopharynx
84. The small pit in the eye which contains the highest concentration of cone cells is the:
- Optic nerve
 - Ora serrata
 - Macula
 - Fovea
 - Saccades
85. What structure is the origin of the blood supply for the retina?
- Optic nerve
 - Optic disk
 - Optic tract
 - Retinal ganglion
 - Ciliary muscles
86. Which of the below is most commonly notably reduced in patients with nyctalopia?
- Rod cells
 - Cone cells
 - Rhodopsin
 - Bipolar neurons
 - Melanin
87. In healthy individuals, which of the following ocular tissues contains blood vessels?
- Cornea
 - Lens
 - Iris
 - A and C
 - All of the above
88. Stimulation of the _____ nervous system results in _____ of the pupil.
- Parasympathetic; constriction
 - Sympathetic; dilation
 - Somatic; dilation
 - Parasympathetic; dilation
 - Sympathetic; constriction

Histology: Use the slide in the **image packet** (pg. 2) to answer the following questions. Point values will be denoted. **(7)**

*Reference: Histology @ Yale

89. **TB#5** Identify **Label A**. **(1)**

90. Identify **Label B**. **(1)**

91. **T/F:** **Label C** contains perilymph. **(1)**

92. **T/F:** The cells at **Label D** do not send signals which are registered as sounds by the brain. **(1)**

93. Which label(s) are components of the spiral organ? **(2)**

94. Which label shown corresponds to the basilar membrane? **(1)**

Histology: Use the slide in the **image packet** (pg. 2) to answer the following questions. Point values will be denoted. **(3)**

*Reference: Histology @ Yale

95. At which region would one expect to find the nuclei for sustentacular cells? **(1)**

- a. Region A
- b. Region B
- c. Region C
- d. Region D
- e. Region E

96. At which region would one expect to find microvilli? **(1)**

- a. Region A
- b. Region C
- c. Region D
- d. Surrounding the nerve bundle
- e. None of the above

97. **T/F:** The cells found in Region B are capable of regenerating via cell division if damaged. **(1)**

Labeling: Complete the provided diagram in the **image packet** (pg. 2). Each label is worth **one** point. **(4)**

98. Label **A**.

99. Label **B**.

100. Label **C**.

101. Label **D**.

Matching: Select the term which corresponds best to the description provided. Some terms may be used more than once, while some may not be used at all. Each match is worth **one** point. **(5)**

A. Myopia

B. Hyperopia

C. Presbyopia

D. Astigmatism

102. **TB#6** Occurs when light is focused in front of the retina.

103. Is associated with unequal refraction of light.

104. Nearby objects are seen as blurry and the eye's ability to accommodate is reduced.

105. Occurs when the eyeball is too short.

106. Is corrected with concave lenses.

Fill-in: Complete the sentences with the most appropriate answer. Be as clear and concise as possible. Each is worth **one** point. **(4)**

107. The phototransduction cascade starts with a conversion of the pigment 11-cis _____ to all-trans _____ [same word 2X].

108. The previous change results in movement of transducin and hydrolysis of cGMP to GMP by phosphodiesterase. GMP closes Na⁺ channels which results in the rod cell undergoing a phenomenon known as _____.

109. The previous conversion results in the rod cell being turned _____.

110. When the rod cell is in the aforementioned state, the _____ to which it is associated with will be activated. This signals to the brain that light is present.

III. Endocrine System

[52 points]

Multiple Choice: Choose the most appropriate answer for the following questions. Each question is worth **one** point. (21)

111. Which of the following is not a steroid hormone?
- a. Oxytocin
 - b. Prolactin
 - c. Testosterone
 - d. A and B
 - e. None of the above
112. Somatostatin is produced by D cells of the stomach and inhibits G cells from secreting gastrin. This is an example of what signaling pathway?
- a. Paracrine
 - b. Autocrine
 - c. Endocrine
 - d. None of the above
113. The thyroid gland cells produce which of the following?
- a. Tetraiodothyronine
 - b. Triiodothyronine
 - c. Calcitonin
 - d. A and B
 - e. All of the above
114. An 11-year-old boy presents to the ER with complaints of polyuria, polydipsia, and polyphagia. His mother claims that her son has lost weight out of nowhere and often complains of being tired. The patient most likely has:
- a. Type 1 Diabetes mellitus
 - b. Congenital iodine deficiency
 - c. Goiter
 - d. Primary hyperaldosteronism
115. There are three major types of pituitary cells. Select the option that matches the pituitary cell type with its correct match.
- a. Acidophil cells - secretes adrenocorticotrophic hormone
 - b. Acidophil cells - secretes prolactin
 - c. Basophilic cells - secretes growth hormone
 - d. Basophilic cells - do not secrete hormones
116. How are steroid and thyroid hormones degraded?
- a. They are degraded by enzymes and excreted in the urine.
 - b. They are inactivated and conjugated in the liver and excreted in bile.
 - c. They undergo proteolytic degradation in the liver.
 - d. They undergo proteolytic degradation in the kidneys.
117. Hormones can also be classified based on their solubility. Which of the following hormones can diffuse through the cell plasma membrane and bind to intracellular receptors?
- a. Steroid hormones
 - b. Thyroid hormones
 - c. Peptide hormones
 - d. A and B
 - e. All of the above

118. What layer of the adrenal cortex produces aldosterone?
- Zona reticularis
 - Zona fasciculata
 - Zona glomerulosa
 - None of the above. The adrenal cortex does not produce aldosterone.
119. When the renin-angiotensin-aldosterone system is activated, which of the following is released?
- Glucocorticoids
 - Androgens
 - Mineralocorticoids
 - A and B
 - All of the above
120. A 34-year-old woman presents to the ER with palpitations, heat intolerance, and bilateral exophthalmos. She also complains that her menstrual cycle is irregular, feelings of anxiety, and double vision which has impaired her driving. The patient most likely has:
- Hashimoto's
 - Polycystic ovary syndrome
 - Hypoglycemia
 - Graves' disease
121. **TB#7** A 22-year-old man goes for a check-up due to anxiety, fatigue, and sleep disturbances. During his check-up, he tells his doctor that he noticed he has more stretch marks, his face seems more flushed, and his skin bruises easily. The doctor also notes that the patient is obese, has a round face, and a hump on his upper central back. The patient most likely has:
- Hashimoto's
 - Myxedema
 - Central diabetes insipidus
 - Cushing's Syndrome
122. Which of the following is not a symptom of Addison disease?
- Increased libido
 - Hyperpigmented skin
 - Hypotension
 - Metabolic acidosis
 - All of the above are symptoms of Addison disease.
123. A 53-year-old man was rushed to the ER after fainting. He also presents with tetany, seizures, and a cardiac arrhythmia. When taking his blood pressure using a cuff, you notice that his hand spasmed. Upon reviewing his past medical history, you note that he recently had anterior neck surgery, seizures, and puffy coarse skin. His labs show that he has a total Ca^{2+} of 6.9 mg/dL. The patient most likely has:
- Hypoglycemia
 - Hyperparathyroidism
 - Hypoparathyroidism
 - Hyperthyroidism
124. Goiters can be classified based on thyroid function. Increased thyroid hormone production is associated with:
- Nontoxic goiter
 - Toxic goiter
 - Hypothyroid goiter
 - All of the above.
125. Hypothyroidism effects include:
- Decreasing sweating
 - Cold skin
 - Decreased prolactin production
 - A and B
 - B and C

126. The kidneys play a role in hormone synthesis. In patients with anemia or patients that experience severe blood loss, what hormone will be produced by the kidneys?
- Calcitriol
 - Prostaglandins
 - Dopamine
 - Erythropoietin
127. At low doses, what hormone helps in the maintenance of renal blood flow and increases sodium excretion via vasodilation of interlobular arteries, afferent arterioles and efferent arterioles?
- Calcitriol
 - Prostaglandins
 - Dopamine
 - Erythropoietin
128. Leptin is a hormone that is mainly produced in adipose tissue. Which of the following statement(s) regarding leptin is correct?
- It induces a feeling of satiety.
 - Lack of sleep increases leptin production.
 - Leptin production is decreased in states of inflammation.
 - B and C
 - None of the above are true.
129. Insulin has effects on adipose tissue. Which of the following is an effect of insulin?
- Increased glucose intake for lipogenesis
 - Decreased lipolysis
 - GLUT4 transporter expression
 - A and B
 - All of the above
130. What is the mode of inheritance for congenital adrenal hyperplasia?
- Autosomal dominant
 - Autosomal recessive
 - X-linked dominant
 - X-linked recessive
 - Mitochondrial
131. 21 β -hydroxylase deficiency, the most common subtype of congenital adrenal hyperplasia, is associated with:
- Primary amenorrhea
 - Hypotension
 - Normal female external genitalia at birth
 - Delayed puberty 17 α -hydroxylase deficiency
 - Male pseudohermaphroditism

T/F: Determine if the following statements are true or false. Each question is worth **one** point. **(10)**

132. The liver has insulin-independent glucose uptake.
133. Sweating, tachycardia, confusion, and somnolence are symptoms of hypoglycemia.
134. Increased adipose tissue can lead to increased conversion of androgen to estrogen.
135. Insulin is derived from an amino acid.
136. Hydrophilic hormones transmit signals by binding to cell membrane receptors.
137. Thyroid hormones (T3 and T4) are derived from cholesterol.

138. Calcitonin is controlled by the hypothalamic-pituitary-thyroid axis.
139. The most common cause of Addison disease in the US is autoimmune adrenalitis.
140. Prostaglandins maintain renal blood flow via vasodilation of efferent arterioles.
141. NSAIDs increase prostaglandin synthesis by blocking COX.

Histology: Use the provided slides in the **image packet** (pg. 3) to answer the following questions. Point values will be denoted. **(13)** *Resource: University of Michigan - Histology*

142. What part of the pituitary gland is shown in **Slide 1?** **(1)**
143. There are wavy, fibrous structures present in **Slide 1**. These structures represent what? **(2)**
144. The structures in **Slide 1** secrete what two hormones? **(2)**
145. State the region of the adrenal gland shown in **Slide 2?** **(1)**
146. What are the cells shown in **Slide 2?** **(1)**
147. **T/F:** Epinephrine is produced by the cells in **Slide 2.** **(1)**
148. **T/F:** Dopamine is produced by the cells in **Slide 2.** **(1)**
149. What gland is shown in **Slide 3?** **(1)**
150. What cells are marked by the green circles in **Slide 3?** **(1)**
151. State two mechanisms of actions of the main secretory protein product released by the cells marked by the blue lines in Slide 3. **(2)**

Endocrine Disruptors: According to the NIH, endocrine disruptors are natural or man-made chemicals that could interfere with hormones and our body's endocrine system. They are not only linked to health problems involving people, but also in wildlife. Answer the following questions regarding endocrine disruptors. Complete sentences are **not** required. **(8)**

152. Which human-made chemical is used to make polycarbonate plastics and epoxy resins?
(1)
- a. DDT
 - b. BPA
 - c. PFAS
153. For DDT, why does body burden increase with age? **(3)**
154. Khalil et al. (2016) demonstrated that 95% of U.S. citizens had PFASs present in their blood stream. What are the two main sources of PFASs exposures? **(2)**
155. **T/F:** PFASs can enter organisms through the skin and with inhaled polluted air (e.g., dust).
(1)
156. **T/F:** BPA is a compound that is an endocrine disruptor and exhibits estrogenic activity. **(1)**

IV. Case Studies

[30 points]

Case 1:

A 27-year-old woman presents to the clinic with complaints of nausea, frequent vomiting, and sensitivity to smell. Notably, she visited the same clinic eight months earlier with a fever of 101°F and a severe, persistent headache. She tested positively for SARS-CoV-2 at that time, and then sent home to recover. A re-test for SARS-CoV-2 at her second visit comes back as negative. The physicians find that she has lost significant weight from a reported inability to eat even foods she used to enjoy. (9)

Sequeria Rodriguez, P., Santana Ortiz, R., & Ortiz-Hernández, E. (2023). PMID: 36691459

157. **TB#8** Diagnose the patient with two issues arising from their SARS-CoV-2 exposure. (3)
158. How long do the diagnoses above **generally** last? In the case of this patient, is it possible for these diagnoses to remain permanent? Briefly explain. (3)
159. **T/F:** The previous diagnoses generally act completely independently of one another. (1)
160. What is the specific term used to describe this patient's report of being unable to eat even familiar, formerly-enjoyable foods due to how they smell? (2)

Case 2:

You are a third-year medical student on a neurology clerkship rotation. A 63-year-old man walks into the clinic. As he walked, you notice that he had a shuffling gait with quickened and shortened steps. His main concerns were a resting tremor in his hand, dystonia, and a stooped posture. He also said that he often feels imbalanced and he has experienced a fall recently. As the patient is sharing his concerns, you note that he has a low degree of facial expression. (14) Reference: AMBOSS

161. Diagnose the patient. (1)
162. Patients with this disorder may also experience depressive symptoms. These symptoms are likely caused by? (4)
163. **Fill-in:** This disease is caused by the progressive _____ degeneration in the _____ and _____, leading to the deficiency of _____ at the receptors of the striatum. (Hint: Some lines do not have a one-word answer.) (4)
164. **TB#9** What is the likely cause of dyskinesia in this disease? (3)
165. **T/F:** Exposure to manganese and low levels of vitamin D are risk factors for this disease. (1)
166. **TB#10 T/F:** There is currently no cure for this disease. (1)

Case 3:

A 42-year-old woman presents to the emergency room and has a chief complaint of fatigue, headaches, muscle weakness, polyuria, and polydipsia. She also has been experiencing constipation and palpitations. The staff took her vital signs and she has a blood pressure of 173/118 mm Hg. The patient later reports that she has been experiencing high blood pressure and is also actively taking three antihypertensive medications. On her exam, she has no peripheral edema. **(7)**

References: AMBOSS; Cobb A, Leslie SW. Primary Hyperaldosteronism. [Updated 2025 May 4]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK539779/>

167. Diagnose the patient. **(1)**
168. What are the two most common etiologies for this disease? **(2)**
169. Patients with this disease are at a heightened cardiovascular risk. State the reason for this risk. **(2)**
170. Surgery is an option for treatment of this disease. If a patient cannot undergo surgery, what treatment is recommended? **(2)**